

From contact to qualification

Extend the RPC Voice Agent — PM Design & Prototyping Assignment

VOIZ Platform · Predixion AI · Time allowed: 2–3 days

Deliverables	Part 1: Product design document + Part 2: Working prototype
Time limit	2–3 days from receipt of this brief
Submission	ZIP or shared links — see submission guidelines on last page
Review format	30-minute walkthrough call after submission
Pass threshold	65 / 100 points across both parts

1. Context

Predixion AI builds VOIZ, a voice agent platform for BFSI clients in India. One live agent makes outbound calls to loan prospects — it confirms the right person is on the line (Right Party Contact), states the offer, and captures interest — all in under 30 seconds, in conversational Hindi (Tier 2/3/4 North India).

The problem: The current agent stops at: *"Are you interested?"* → handoff to a human sales rep. Human reps report spending 40–60% of every first call re-qualifying leads the voice agent already spoke to. Basic parameters — loan amount needed, purpose, employment type — were never captured, so the rep starts from scratch.

Your brief: Design and prototype a qualification layer that sits between interest confirmation and human handoff.
The agent must collect 2–3 qualifying data points before the call ends — without exceeding a 55-second total call duration, and without alienating Tier 2/3 customers who distrust long automated calls.

Hard constraints you must design within

- **Total call duration cap:** 55 seconds (was 30s for RPC-only; you have 25 extra seconds for qualification)
- **Language:** Hindi only, conversational Tier 2/3/4 register — no corporate or translated-English phrasing
- **LLM call budget:** max 4 per call (2 already used for RPC + interest check; you have 2 to add)
- **STT accuracy:** Deepgram Nova-2, Hindi. Expect ~15% WER on noisy or low-signal calls — design for it
- **Cost sensitivity:** Client is cost-conscious. Every additional INR 0.10/call must be explicitly justified
- **Data restriction:** Do not collect sensitive data via voice (Aadhaar, PAN, exact income figures)

2. Part 1 · Product Design Document

Write a single structured document (PDF or Notion export) covering all four sections below. Write in prose — not slides. Aim for clarity over volume.

2.1 Problem framing & success definition

Target length: ~1 page

1. Define what a 'qualified lead' means in this context — from the client's perspective (what does their sales team actually need?) and the customer's perspective (what makes this call feel worth their time?).
2. Write exactly 3 success metrics with targets you'd commit to for the first 1,000 calls. For each metric, state: what it measures, how it's calculated, what the target is, and why you chose it over alternatives you considered.
3. Identify the single biggest risk in your design. State how you'd detect it early and what your rollback condition is.

Weak submissions pick vanity metrics (call completion rate, NPS). Strong ones tie metrics directly to the problem stated above — reduction in rep re-qualification time, improvement in lead-to-application conversion, qualification drop-off rate.

2.2 Extended call flow design

Target: Full FSM diagram + annotated script. Tool of choice — draw.io, Miro, FigJam, or hand-drawn and photographed.

4. Design the qualification sequence. Choose 2–3 questions to ask, in a specific order. Justify: why these questions, why this order, and — critically — what you dropped and why.
5. Draw the complete FSM from call start through CRM handoff. All states, all transitions, all exit paths must be shown. Partial diagrams will be marked down.
6. **For each new question state, specify:** (a) the exact Hindi script line the agent speaks, (b) the intent labels the classifier is trained to return, (c) what happens on each label including UNCLEAR, (d) the timeout behaviour if the customer doesn't respond.
7. **Define the handoff payload:** What structured data object is sent to the CRM and surfaced to the sales rep? Include field names, data types, and what value is written when data was not captured.

You cannot fit all three of — loan amount, purpose, employment type, tenure — within 25 extra seconds. The brief requires you to pick 2–3 and justify the cut. Attempting to include all four will be marked as a prioritisation failure regardless of how good the rest of the design is.

2.3 Edge case & failure mode analysis

Target: Table or structured list, minimum 6 edge cases.

Map each edge case across these four columns:

Edge case	Agent behaviour	What gets logged	Human follow-up?
Customer gives out-of-range or nonsense answer	Define	Define	Define
Background noise — STT returns garbage	Define	Define	Define
Customer switches from Hindi to English mid-call	Define	Define	Define
Customer asks 'who are you calling from?'	Define	Define	Define
Customer gives UNCLEAR twice in a row on same question	Define	Define	Define
Hard timeout fires mid-qualification question	Define	Define	Define
[Your additional cases]	...	...	...

Also state: at what qualification drop-off rate would you recommend disabling the qualification layer entirely and reverting to pure RPC handoff? Provide a threshold and the business logic behind it.

2.4 Cost & tradeoff analysis

Target: ~0.5 page. Show your working.

- 8. Estimate the incremental cost per call added by your qualification design. Break it down by: extra STT seconds, additional LLM classifier calls, additional TTS characters. Use realistic per-unit figures — research Deepgram, OpenAI, and Sarvam/ElevenLabs pricing.
- 9. Calculate the break-even point: at what improvement in lead-to-application conversion rate does the added cost per call pay for itself? State your assumptions explicitly.
- 10. If the client asks you to reduce the incremental cost by 40%, what do you cut first and what quality tradeoff does that introduce?

Reference rates (as of mid-2024): Deepgram Nova-2 ~\$0.0043/min, GPT-4o-mini ~\$0.15/1M input tokens, Sarvam Bulbul-v2 ~\$0.006/1k characters. Verify current rates and flag if different.

3. Part 2 · Prototype

Build one of the three options below. Read all three before choosing. Each tests different skills — pick the one that plays to your strengths. Working beats polished: a functional prototype with rough UI beats a beautiful mockup that doesn't work.

Option A · Conversation simulator (Recommended)

Recommended for most candidates

Build a browser-based tool where a reviewer can step through your designed call flow by typing customer responses. The tool must show:

- **Current state name** (e.g. QUAL_LOAN_AMOUNT)
- **Agent TTS line** for that state — the exact Hindi script you designed
- **Classifier label produced** from the typed input (you may simulate this with keyword matching or a real LLM API call)
- **Next state** the FSM transitions to, and why
- **Live call timer** counting up from 0 — must fire a visual alert at 55 seconds
- **Call outcome summary** at call end: all captured data fields and their values

No telephony required — keyboard input stands in for STT. Stack: plain HTML/JS, React, or any framework you know. Prioritise functionality over styling.

Tests: FSM implementation thinking, edge case coverage, ability to spec a tool a dev could build from.

Option B · Live voice prototype (Advanced)

For candidates with voice AI tool experience

Build an end-to-end callable prototype using Voiceflow, Bland AI, Vapi + n8n, or equivalent. Must support actual voice input/output in Hindi.

- Submit a callable phone number or Voiceflow share link
- Include a Loom recording (max 5 minutes): one full happy-path walkthrough and one failure path (e.g. UNCLEAR twice on a qualification question)
- The prototype must enforce the 55-second duration cap — demonstrate this in your Loom

Tests: Hands-on tool proficiency, real constraint exposure (timing, STT accuracy), initiative to build something live.

Option C · Ops dashboard mockup (Alternative)

For candidates whose strength is operator / data product thinking

Design the real-time dashboard a collections ops manager uses to monitor live qualification call outcomes. Must include all of:

- **Call outcome distribution:** INTERESTED / NOT_INTERESTED / WRONG_PARTY / TIMEOUT / ERROR — today vs. 7-day trend
- **Qualification funnel:** drop-off at each qualification state — where are customers falling out?
- **Live calls feed:** currently active calls with state, duration, and last classifier output
- **Breach alerts surface:** calls that exceeded duration cap, calls that hit LLM cap, calls with repeated UNCLEAR
- **One drill-down view:** click a completed call → see full state transcript with timing

Figma, Framer, or interactive HTML. Include a 2-minute Loom walkthrough explaining your design decisions — what you prioritised, what you left out, and what you'd add in v2.

Tests: Operator empathy, data product sense, ability to translate call engine outputs into actionable visibility.

4. Submission Guidelines

Read this section in full before you start work — submission format requirements are enforced.
Non-compliant submissions are returned without review.

4.1 What to submit

Item	Format	Required?
Part 1 — Design document	Single PDF or Notion share link (public view). No Google Docs (formatting breaks across machines).	Yes
Part 2 — Prototype	Option A/C: ZIP containing index.html + assets, or a hosted link (Vercel, Netlify, GitHub Pages). Option B: Callable number or Voiceflow link + Loom URL.	Yes
README.md	Plain text file, max 200 words. See required structure below.	Yes
Loom walkthrough	Required for Option B and Option C. Max 5 minutes. No scripts — talk through it naturally.	Option B/C only

4.2 README.md — required structure

Your README must contain exactly these three sections and nothing else:

```
README.md
## What I built
One paragraph. State which prototype option you chose and what it does.
If it requires setup steps, list them here.

## What I'd do differently with more time
One paragraph. Be specific — 'improve the UI' is not an answer.
Name the design decision you're least confident in and why.

## One assumption I wish I could validate
One sentence. The most important unknown your design depends on.
```

We read the README before opening the prototype. A vague or missing README signals poor communication skills regardless of how strong the rest of the work is. This section is worth 10 points.

4.3 How to send it

- **Email subject:** PM Assignment — [Your Name] — VOIZ RPC Agent
- **Send to:** juhi@predixion.ai (CC the person who sent you this brief)
- **ZIP naming:** pm_assignment_[yourname].zip
- **If submitting hosted links:** paste all URLs directly in the email body — do not put them only inside the ZIP
- **File size limit:** 25 MB for ZIP attachments. Use hosted links for anything larger.

4.4 Timeline

Day	Suggested focus
Day 1	Read brief fully. Research Hindi voice UX patterns and Tier 2/3 customer behaviour. Define your 3 metrics. Decide which qualifying questions to include and sketch the FSM on paper before touching any tool.
Day 2	Finalise FSM and scripts. Write Part 1 document sections 2.1–2.4. Begin prototype build.
Day 3 (if needed)	Complete prototype. Write README. Review your FSM one more time — check every exit path. Submit before 11:59 PM IST.

A clean submission on Day 2 is better than an overloaded one on Day 3. We penalise scope creep, not conciseness.

4.5 What you may and may not use

Permitted

- Any AI coding assistant (Copilot, Cursor, Claude) for prototype code
- Any diagramming tool for the FSM (draw.io, Miro, FigJam, Lucidchart, pen and paper)
- Any no-code voice tool for Option B (Voiceflow, Bland AI, Vapi)
- Public documentation, pricing pages, and blog posts for research

Not permitted

- Generating Part 1 prose entirely via AI — we will ask you to defend every design decision in the review call
- Copying another candidate's submission — both submissions will be disqualified
- Adding team members or asking others to build the prototype for you

In the review call, you will be asked to modify your FSM live. If you can't explain why each transition exists, the score reflects it.

5. Evaluation Rubric

Shared with candidates in full — no hidden criteria.

Area	What we look for	Points
Problem framing	Metrics are specific, measurable, and tied directly to the stated problem — not vanity metrics. Risk identification is honest, not defensive.	15
FSM & call flow design	State machine is complete — no missing transitions or undefined exits. Hindi scripts are natural Tier 2/3 register, not translated English. Prioritisation decision is explicitly stated and logically argued.	25

Area	What we look for	Points
Edge case coverage	Handles the six mandatory cases plus any others the design introduces. UNCLEAR handling is consistent. Rollback threshold is defined with business logic, not just a number.	15
Cost & tradeoff analysis	Numbers are grounded — sources cited. Break-even logic is sound. Candidate demonstrates understanding of the cost levers in a voice AI stack.	10
Prototype quality	Works without hand-holding. Happy path and at least one failure path are demonstrable. Code or design decisions are defensible in the walkthrough call.	25
README & communication	Concise, honest, and specific. Identifies a real unknown rather than hedging everything. No marketing language. Completed correctly as specified.	10

Pass threshold: 65 / 100. Candidates below 65 are not progressed regardless of prototype quality or seniority.

Automatic disqualifiers: Missing README, broken prototype with no setup instructions, FSM with undefined exit paths, or inability to explain design decisions in the review call.

6. Review Call Format

The review call is 30 minutes. Structure:

Segment	Duration	What happens
Prototype walkthrough	10 min	You walk us through the prototype end-to-end — happy path, then one failure path. No slides. Share screen and talk through it.
FSM live edit	10 min	We describe a new edge case. You modify your FSM diagram live on screen. We're evaluating how you think under constraint, not whether the final answer is perfect.
Design decisions Q&A	8 min	We ask why you chose your 2–3 qualifying questions over alternatives. Be prepared to defend the cut decisions specifically.
Your questions	2 min	Ask us anything about the role, the product, or the team.

Come prepared to say 'I don't know' about things you didn't design — that's fine.
Come unprepared to explain decisions you did make — that is not fine.